DEEP LEARNING: ADVANCED MODELS AND METHODS

Knowledge Distillation for Mobile Action Recognition

Compressing 3D ResNet-50 into MobileNet3D: Optimization, Attention Transfer, and Cross-Frame Distillation

PROJECT DETAILS

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INTRODUCTION

Project Objectives & The Capacity Gap

THE COMPRESSION CHALLENGE

• Mobile Action Recognition: Video processing is computationally heavy, demanding compact and fast models for mobile or edge deployment.

• Target: Compress a heavy 3D ResNet-50 Teacher (~128 MB, Kinetics-400 pretrained) into an ultra-lightweight MobileNet3D Student (~9.5 MB) trained from scratch.

• The Capacity Gap: MobileNet3D trained from scratch from raw labels reaches a performance ceiling of 59.48% Test Top-1, creating a massive 29.34% gap with the Teacher (88.82%).

• The Solution: Employ Knowledge Distillation (KD) to transfer the Teacher's dark knowledge and structural representations to close this performance gap.

ARCHITECTURAL COMPARISON

Model	Size	Top-1	Top-5
Teacher (3D R50)	128.0 MB	88.82%	98.18%
Student Baseline	9.5 MB	59.48%	82.77%

The Teacher is ~13.5x larger than the Student baseline, presenting a clear compression opportunity via logit and attention alignment.

METHODOLOGY

Experimental Setup & Group-Aware Dataset Split

TRAINING HYPERPARAMETERS & CONFIG

- Optimizer: AdamW with weight decay of 0.01.
- LR Scheduler: Cosine annealing with base learning rate of 0.0005 for students, 0.0007 for teacher.
- Training Budget: 60 epochs (with 10-epoch warmup for KD/AT where only Cross-Entropy loss is active).
- Regularization: AMP (Automatic Mixed Precision) enabled, Label Smoothing of 0.1.
- Data Augmentation (Light): Short-side resize to 128, crop to 112x112, random horizontal flip, color jitter (0.1), random erasing (0.05), max temporal stride 1.

GROUP-AWARE SPLIT DESIGN

- Motivation: UCF-101 contains clips extracted from the same original source videos. Standard random splitting results in data leakage, creating overly optimistic, fake validation metrics.
- **Smart Division: The training set was split in a wise, group-aware manner based on source video IDs (extracted via regex, e.g. '_gXX_') to obtain an internal evaluation set which was otherwise absent.**
- Impact: Evaluation set metrics dropped from fittitiouly high values to a realistic 60-65% validation range, exposing the true generalization gap and ensuring reliable hyperparameter tuning.

Logit-Based Knowledge Distillation Results

LOGIT ALIGNMENT METHODOLOGY

- Loss Formulation: Combines standard Cross-Entropy (hard labels) with KL Divergence on soft targets scaled by temperature T.
- Hyperparameters: Best standard baseline configuration uses temperature T = 8.0, alpha = 0.7, batch size 16.
- **Warmup Effect: An initial 10-epoch classification warmup using pure ground truth is crucial to stabilize Student weights before introducing teacher soft targets.**
- Parameter Overhead: +0% extra parameters. The distilled student maintains the exact 9.5 MB size and low latency of the baseline.

QUANTITATIVE PERFORMANCE COMPARISON

Model	Val Top-1	Test Top-1	Test Top-5	Delta
Teacher (3D R50)	92.36%	88.82%	98.18%	Upper Bound
Student Baseline	59.18%	59.48%	82.77%	Reference
KD Student (T=8)	64.57%	**64.31%**	87.68%	**+4.83%**

Knowledge Distillation yields a significant +4.83% absolute improvement in Test Top-1 and +4.91% in Top-5, without modifying the Student architecture.

KNOWLEDGE DISTILLATION

Temperature Scaling Ablation Study

TEMPERATURE ABLATION SWEEP (ALPHA = 0.7)

Temp (T)	Train Acc	Val Top-1	Test Top-1	Delta vs BL
T = 1	98.44%	62.75%	62.91%	+3.43%
T = 5	95.87%	63.12%	62.07%	+2.59%
T = 8	97.59%	64.57%	64.31%	+4.83%
T = 10	98.28%	64.85%	64.47%	+4.99%
T = 20 (Best)	**99.05%**	**65.18%**	**65.85%**	**+6.37%**

KEY TEMPERATURE TRENDS & FINDINGS

• Monotonic Increase: Accuracy shows a clear upward trend with temperature, peaking at T = 20 with a massive +6.37% gain over baseline.

• Softening Probability Distributions: With T=1, the teacher's soft targets are too peaked, behaving like standard hard labels. This makes it difficult for the student to capture relationships between classes.

• Bridging the Capacity Gap: Setting T=20 smooths out the distributions, revealing the 'dark knowledge' (inter-class similarity structures) that a lower-capacity student is capable of mimicking.

ADVANCED KD METHODS

Attention Transfer: Debugging & Results

THE 'KEY 5' BUG & FEATURE PAIRING

• Concept: Align intermediate spatial and temporal attention maps (L2-normalized activations) from matching network stages.

• The Bug: Early attempts targeted block 5 of the Teacher (ResNet-50 classification head, 2D). This generated constant 1.0 attention maps, silently nullifying the loss.

• Correction: Mapped stages with comparable spatial dimensions: `[2,3,4]` of ResNet to `[2,4,6]` of MobileNet.

• Configs: Symmetric (spatial beta=0.05, temporal beta=0.05) and Temporal-Only (spatial beta=0.0, temporal beta=0.10).

PERFORMANCE REGRESSION & ANALYSIS

Configuration	Val Top-1	Delta vs KD (T=8)
KD (T=8, no AT)	64.57%	Reference
AT Symmetric	58.25%	**-6.32%**
AT Temporal-Only	59.04%	**-5.53%**

Why did Attention Transfer fail?

1. Capacity Gap: Standard 3D convolutions (Teacher) produce rich, high-dimensional spatial maps. MobileNet3D's depthwise separable convolutions are too limited to mimic them, causing learning collapse.

2. Over-Regularization: Forcing intermediate layers to strictly align acts as an excessive constraint, preventing the Student from finding alternative solutions.

ADVANCED KD METHODS

Born-Again Networks: Generational Collapse

ITERATIVE SELF-DISTILLATION SETUP

• Method: Train a student model from a teacher. In the next generation, this student becomes the new teacher, distilling its knowledge to a structurally identical student.

• Homogeneous Architecture: MobileNet3D (Teacher) to MobileNet3D (Student). Spatial features match perfectly without interpolation.

• Initial Target (Gen 0): The best-performing logit-distilled student (KD T=8, 64.31% Test Top-1) serves as the starting point.

• Configuration: Temperature T = 8.0, alpha = 0.7, optimizer AdamW, cosine scheduler, light augmentation.

THE COLLAPSE PHENOMENON

Generation	Teacher	Test Top-1	Delta vs Gen 0
Gen 0	ResNet-50	64.31%	Reference
Gen 1	Gen 0 Student	60.98%	**-3.33%**
Gen 2	Gen 1 Student	60.43%	**-3.88%**
Gen 3	Gen 2 Student	59.21%	**-5.10%**

Why did the generations collapse?

1. Uncalibrated Temperature: T=8.0 was tuned for the high-confidence ResNet-50. When applied to the less confident MobileNet3D (64%), it flattened probabilities into near-white noise.

2. Error Propagation: Without an oracle (ground truth teacher), systematic mistakes and artifacts accumulate and propagate across generations, leading to overfitting on errors.

Latent Space Analysis via t-SNE Projections

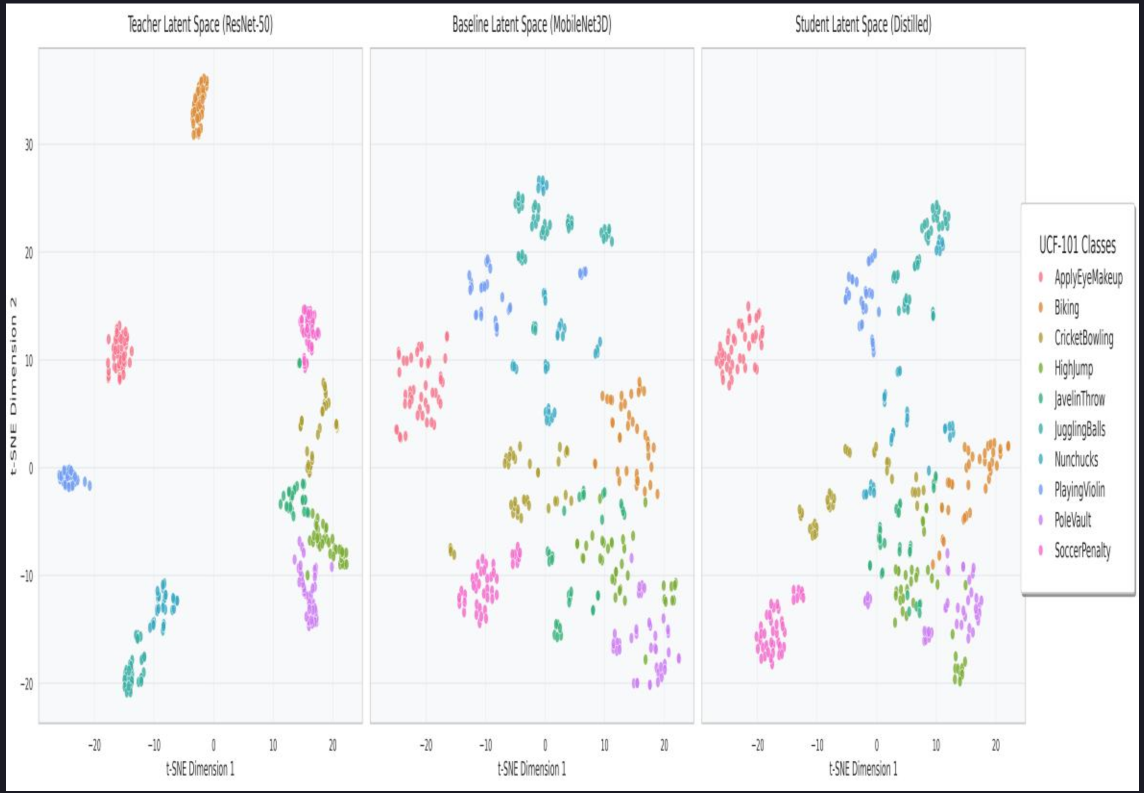
VISUALIZING LEARNED EMBEDDINGS

• Method: Apply t-SNE dimensionality reduction to pre-classifier embeddings across 10 randomly selected classes.

• Teacher (ResNet-50): Left panel shows highly compact, well-separated, and distinct semantic clusters.

• Student Baseline: Middle panel displays fuzzy cluster boundaries, significant class overlap, and mixed clusters.

• Distilled Student (KD T=8): Right panel shows dramatically improved cluster definitions and separation, validating the success of structural knowledge transfer.



CROSS-FRAME DISTILLATION

Cross-Frame Distillation for Edge Deployment

EFFICIENCY VS ACCURACY GOALS

• Deployment Challenge: Processing 24 frames per video clip is computationally expensive, especially on resource-constrained CPUs and older edge GPUs.

• Cross-Frame Distillation: Train a Model B using only 16 frames per clip, representing a 33.3% theoretical reduction in temporal FLOPs compared to the 24-frame Model A.

• Core Inquiry: Verify if this 33.3% computational savings translates into actual wall-clock execution speedups on hardware (GPU and CPU).

BENCHMARK METRICS & METHODOLOGY

• Dataset: Official UCF-101 test set, containing 3,783 real video clips.

• Measurement: Focuses solely on forward inference time (model(clips)) to isolate network speed from disk I/O.

• GPU Sync: Uses PyTorch CUDA synchronization to ensure asynchronous kernels are fully completed before measuring times.

• CPU Sampling: CPU benchmarks are sub-sampled (50 batches for BS=1, 20 batches for BS=8) to avoid excessively long test runs while retaining statistical accuracy.

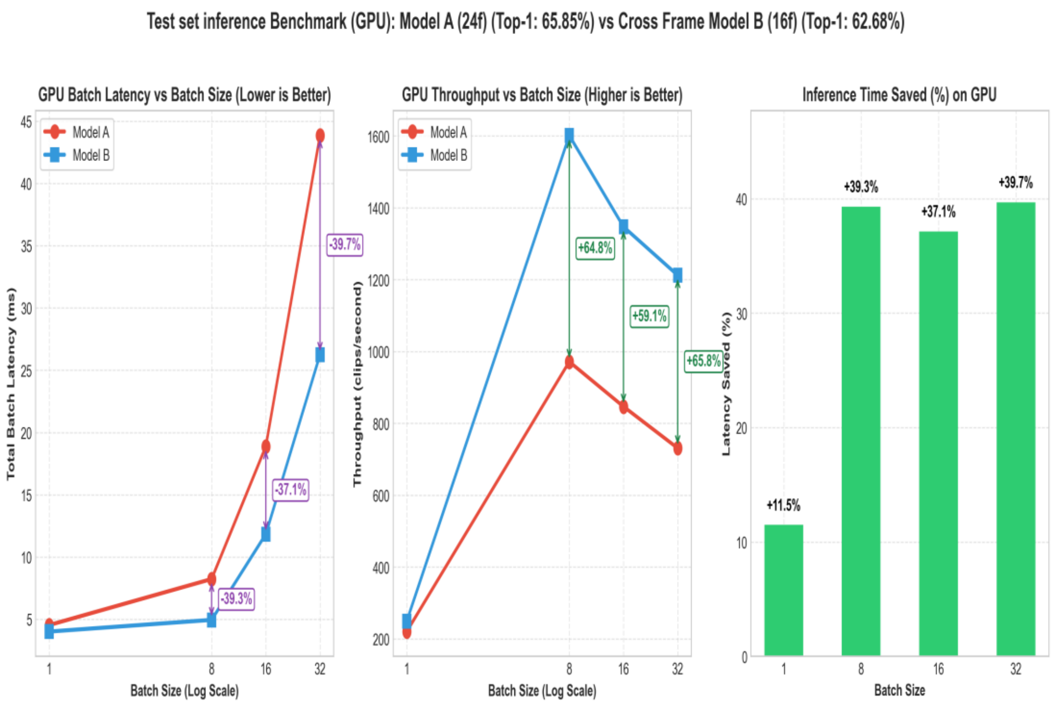
GPU Inference Benchmark: NVIDIA L40S

GPU BENCHMARK (MS PER CLIP)

Batch Size	Model A (24f)	Model B (16f)	Speedup	Saved Time
BS = 1	4.54 ms	4.02 ms	1.13x	11.49%
BS = 8	1.03 ms	0.62 ms	**1.65x**	**39.32%**
BS = 16	1.18 ms	0.74 ms	**1.59x**	**37.15%**
BS = 32	1.37 ms	0.82 ms	**1.66x**	**39.69%**

Key GPU Findings:

- GPU Saturation: At BS >= 8, GPU is fully saturated, yielding a massive 37%-40% speedup. This exceeds the linear 33.3% frame reduction due to better alignment of 16-frame blocks on Tensor Cores.
- PCIe Overhead: At BS = 1, PCIe bus overhead dominates, limiting speedup to 1.13x.



CROSS-FRAME DISTILLATION

CPU Inference Benchmark: Host Cluster

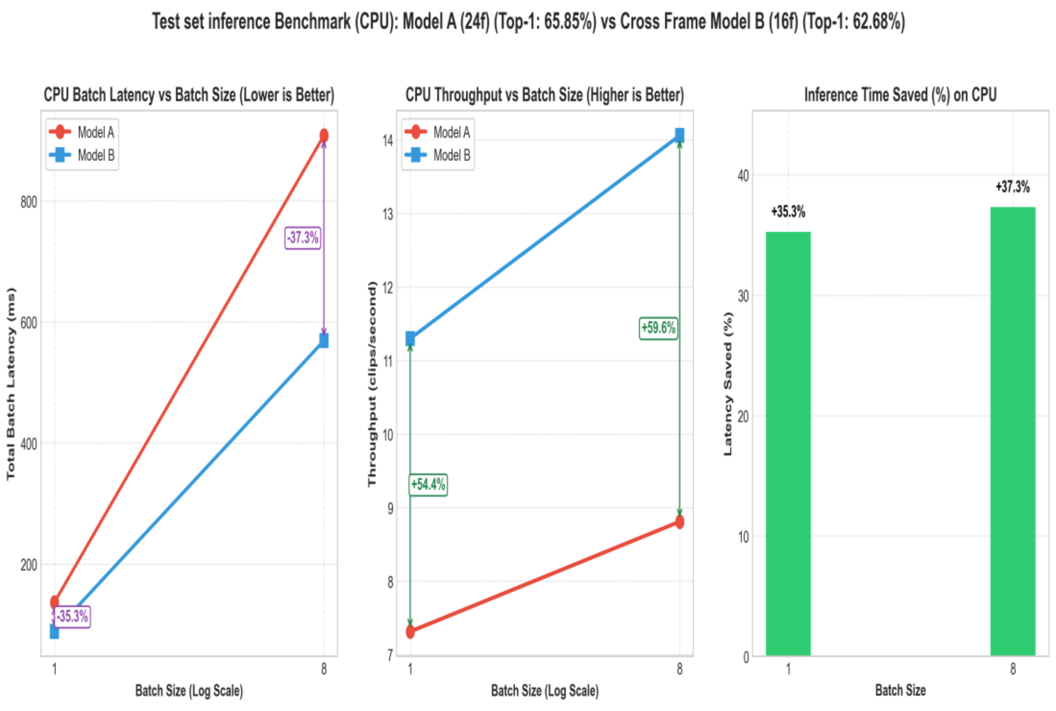
CPU BENCHMARK (MS PER CLIP)

Batch Size	Model A (24f)	Model B (16f)	Speedup	Saved Time
BS = 1	136.70 ms	88.51 ms	**1.54x**	**35.26%**
BS = 8	113.51 ms	71.13 ms	**1.60x**	**37.33%**

Key CPU Findings:

• FLOPs-Dominated Speedup: Without the PCIe bus bottleneck, CPU speedup closely reflects the theoretical temporal FLOP reduction.

• Edge Suitability: Reducing single-clip latency (BS=1) from 136.7 ms to 88.5 ms (below the 100 ms threshold) makes the 16f model highly suitable for real-time edge processing.



CONCLUSIONS

Conclusions & Speed-Accuracy Trade-off

THE SPEED-ACCURACY TRADE-OFF

• Model A (24f, Best KD T=20): Achieves 65.85% Test Top-1 (a substantial +6.37% gain over the baseline scratch student).

• Model B (16f, Cross-Frame KD): Reaches 62.68% Test Top-1.

• Trade-off Evaluation: Saving 37%-40% in latency/FLOPs costs only 3.17% in Test Top-1 compared to Model A.

• Net Improvement: Model B (16f) still outperforms the original Model A baseline from scratch (59.48%) by a solid +3.20% while executing significantly faster.

KEY SUMMARY TAKEAWAYS

• Logit-Based KD is a powerful and zero-overhead method. Higher temperatures (T=20) are critical to soften representations and bridge large capacity gaps.

• Attention Transfer is limited by architectural heterogeneity (standard vs depthwise convolutions) and risk of over-regularization.

• Self-Distillation (BAN) is highly sensitive to temperature and prone to generational collapse when the teacher is not an oracle.

• Cross-Frame Distillation provides an outstanding Pareto-optimal trade-off for practical mobile/edge real-time deployments.